

Linear Phase Planes and Stability

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Definition: Planar linear system

A **homogeneous planar linear system** has the form

$$\dot{\mathbf{x}} = A\mathbf{x}, \quad A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad A \in \mathbb{R}^{2 \times 2}, \quad \mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2.$$

Thus $a, b, c, d \in \mathbb{R}$.

Definition: Phase plane

The **phase plane** of a planar system is its two-dimensional state space.

Recall: Trajectory

A **trajectory** is the image of one solution in the phase plane.

Recall: Phase portrait

A **phase portrait** is a diagram of the qualitatively distinct trajectories and equilibria in the phase plane, with the direction of increasing time marked on each non-equilibrium trajectory.

Observation: Equilibrium at the origin

The origin is an equilibrium of every homogeneous linear system because

$$A\mathbf{0} = \mathbf{0}.$$

Definition: Invariant set

Let Φ be a possibly local flow. A set M is **invariant** if, for every $\mathbf{x}_0 \in M$ and every t in the maximal existence interval of the solution through $\mathbf{x}_0$,

$$\Phi(t, \mathbf{x}_0) \in M.$$

Definition: Stable set

For an equilibrium $\mathbf{x}^*$, its **stable set** is

$$W^s(\mathbf{x}^*) = \{\mathbf{x}_0 : \Phi_t(\mathbf{x}_0) \rightarrow \mathbf{x}^* \text{ as } t \rightarrow \infty\},$$

where the solution through $\mathbf{x}_0$ is required to exist for every $t \geq 0$.

Definition: Unstable set

For an equilibrium $\mathbf{x}^*$, its **unstable set** is

$$W^u(\mathbf{x}^*) = \{\mathbf{x}_0 : \Phi_t(\mathbf{x}_0) \rightarrow \mathbf{x}^* \text{ as } t \rightarrow -\infty\}.$$

Here the solution through $\mathbf{x}_0$ is required to exist for every $t \leq 0$.

Definition: Stable and unstable manifolds

When $W^s(\mathbf{x}^*)$ is a single point or a smooth curve or surface, it is called the **stable manifold** of $\mathbf{x}^*$. When $W^u(\mathbf{x}^*)$ has one of these forms, it is called the **unstable manifold** of $\mathbf{x}^*$.

Proposition: Superposition

Let $A \in \mathbb{R}^{2 \times 2}$. If $\mathbf{x}_1$ and $\mathbf{x}_2$ solve $\dot{\mathbf{x}} = A\mathbf{x}$, then

$$c_1\mathbf{x}_1 + c_2\mathbf{x}_2$$

is a solution for all $c_1, c_2 \in \mathbb{R}$.

Proof.

$$\frac{d}{dt}(c_1\mathbf{x}_1 + c_2\mathbf{x}_2) = c_1A\mathbf{x}_1 + c_2A\mathbf{x}_2 = A(c_1\mathbf{x}_1 + c_2\mathbf{x}_2).$$

Question: What is the phase portrait of a harmonic oscillator?

Let $m, k > 0$ and consider

$$m\ddot{x} + kx = 0.$$

Find a planar system, its conserved energy, its trajectories, and its period.

Solution. Define

$$v = \dot{x}, \quad \omega = \sqrt{\frac{k}{m}}.$$

Then

$$\dot{x} = v, \quad \dot{v} = -\omega^2 x.$$

The energy

$$E(x, v) = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$$

satisfies

$$\dot{E} = mv\dot{v} + kx\dot{x} = mv(-\omega^2 x) + kxv = 0.$$

Thus each nonzero trajectory lies on an ellipse

$$mv^2 + kx^2 = 2E_0, \quad E_0 > 0.$$

Question: What is the phase portrait of a harmonic oscillator? (continued)

The explicit solution with initial state (x_0, v_0) is

$$x(t) = x_0 \cos \omega t + \frac{v_0}{\omega} \sin \omega t,$$

$$v(t) = -\omega x_0 \sin \omega t + v_0 \cos \omega t.$$

Every nonzero solution has period

$$T = \frac{2\pi}{\omega}.$$

At $(x, 0)$ with $x > 0, \dot{v} < 0$; therefore the ellipses are traversed clockwise in the (x, v) plane.

Question: What is one complete physical oscillation?

Let $m = 2, k = 8, E_0 = 4$, and $(x_0, v_0) = (1, 0)$. Find the orbit and mark the states after each quarter-period.

Solution. Since $\omega = 2$,

$$v^2 + 4x^2 = 4, \quad x(t) = \cos 2t, \quad v(t) = -2 \sin 2t, \quad T = \pi.$$

Hence

t	$x(t)$	$v(t)$
0	1	0
$\pi/4$	0	-2
$\pi/2$	-1	0
$3\pi/4$	0	2
π	1	0

and the state moves clockwise around the ellipse.

Recall: Attracting equilibrium

Let $\mathbf{x}^*$ be an equilibrium of $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$. It is **attracting** if there exists $\delta > 0$ such that every solution with $\|\mathbf{x}_0 - \mathbf{x}^*\| < \delta$ exists for all $t \geq 0$ and

$$\Phi_t(\mathbf{x}_0) \rightarrow \mathbf{x}^* \quad (t \rightarrow \infty).$$

Recall: Lyapunov stability

An equilibrium $\mathbf{x}^*$ is **Lyapunov stable** if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that every solution with $\|\mathbf{x}_0 - \mathbf{x}^*\| < \delta$ exists for all $t \geq 0$ and

$$\|\Phi_t(\mathbf{x}_0) - \mathbf{x}^*\| < \varepsilon \quad \text{for all } t \geq 0.$$

Recall: Asymptotic stability

An equilibrium is **asymptotically stable** if it is both attracting and Lyapunov stable.

Definition: Neutral stability

An equilibrium is **neutrally stable** if it is Lyapunov stable but not attracting.

Recall: Instability

An equilibrium is **unstable** if it is not Lyapunov stable.

Visualization: Convergence and confinement

property	geometric content
attracting	nearby solutions converge to the equilibrium
Lyapunov stable	nearby solutions remain nearby
asymptotically stable	both convergence and confinement

For the harmonic oscillator, nearby energy ellipses remain nearby but never shrink:

$$\boxed{\text{Lyapunov stable}} + \boxed{\text{not attracting}} = \boxed{\text{neutrally stable}}.$$

Question: How does one diagonal family change its phase portrait?

For

$$\dot{x} = ax, \quad \dot{y} = -y,$$

determine the stability of the origin and the generic forward asymptotic behavior for every $a \in \mathbb{R}$.

Solution.

$$x(t) = x_0 e^{at}, \quad y(t) = y_0 e^{-t}.$$

Therefore

a	stability and geometry	generic forward behavior $x_0 y_0 \neq 0$
$a < -1$	asymptotically stable	approach tangent to the y -axis
$a = -1$	asymptotically stable; each nonzero trajectory is a ray	approach along the initial ray
$-1 < a < 0$	asymptotically stable	approach tangent to the x -axis
$a = 0$	$y = 0$ is a line of equilibria; $\mathbf{0}$ is neutrally stable	vertical approach to $(x_0, 0)$
$a > 0$	$\mathbf{0}$ is unstable	escape tangent to the x -axis

For $a > 0$,

$$W^s(\mathbf{0}) = \{(0, y) : y \in \mathbb{R}\}, \quad W^u(\mathbf{0}) = \{(x, 0) : x \in \mathbb{R}\}.$$

Question: How can one prove stability directly?

For

$$\dot{x} = -2x, \quad \dot{y} = -y,$$

prove Lyapunov stability and attraction by estimating the Euclidean norm.

Solution. The solution satisfies

$$\|(x(t), y(t))\|^2 = x_0^2 e^{-4t} + y_0^2 e^{-2t} \leq e^{-2t} \|(x_0, y_0)\|^2.$$

Taking square roots gives

$$\|(x(t), y(t))\| \leq e^{-t} \|(x_0, y_0)\|.$$

For every $\varepsilon > 0$, an initial norm less than ε stays less than ε for every $t \geq 0$, proving Lyapunov stability. The same bound tends to zero, proving attraction.

Question: Can attraction occur without Lyapunov stability?

On S^1 , analyze

$$\dot{\theta} = 1 - \cos \theta.$$

Solution. The unique fixed point is $[0]$. Since

$$1 - \cos \theta > 0$$

for $\theta \notin 2\pi\mathbb{Z}$, every nonfixed trajectory moves counterclockwise and passes through every intermediate angle. In the representative interval $[0, 2\pi)$, its angle approaches 2π , which represents $[0]$. Hence $[0]$ is attracting.

Initial points with $\theta_0 > 0$ arbitrarily small must pass through the opposite point $[\pi]$. Thus arbitrarily close initial states leave the fixed semicircle centered at $[0]$, disproving Lyapunov stability.

Definition: Involution

Let $U \subseteq \mathbb{R}^2$. A map $R : U \rightarrow U$ is an **involution** if

$$R(R(\mathbf{z})) = \mathbf{z}$$

for every $\mathbf{z} \in U$.

Definition: Reversing symmetry

Let $U \subseteq \mathbb{R}^2$ be open, and let $R : U \rightarrow U$ be a differentiable involution. For a vector field $\mathbf{F} : U \rightarrow \mathbb{R}^2$, the map R is a **reversing symmetry** of

$$\dot{\mathbf{z}} = \mathbf{F}(\mathbf{z})$$

if

$$\mathbf{F}(R(\mathbf{z})) = -DR(\mathbf{z})\mathbf{F}(\mathbf{z}).$$

Here $DR(\mathbf{z})$ is the derivative matrix of R . For $\mathbf{z} = (x, y)$ and $R = (R_1, R_2)$,

$$DR = \begin{pmatrix} \partial R_1 / \partial x & \partial R_1 / \partial y \\ \partial R_2 / \partial x & \partial R_2 / \partial y \end{pmatrix}.$$

Observation: Time reversal of a solution

If R is a reversing symmetry and $\mathbf{z}(t)$ is a solution, then $R(\mathbf{z}(-t))$ is also a solution. Indeed, the chain rule gives

$$\frac{d}{dt}R(\mathbf{z}(-t)) = -DR(\mathbf{z}(-t))\mathbf{F}(\mathbf{z}(-t)) = \mathbf{F}(R(\mathbf{z}(-t))).$$

Proposition: Closure of the harmonic-oscillator orbit by symmetry

For

$$\dot{x} = v, \quad \dot{v} = -x,$$

define

$$S_x(x, v) = (x, -v), \quad S_v(x, v) = (-x, v).$$

Both maps are reversing symmetries. Suppose a trajectory begins at

$$\mathbf{z}(0) = (0, -v_0), \quad v_0 > 0,$$

and first meets the x -axis at

$$\mathbf{z}(t_1) = (x_1, 0), \quad x_1 < 0.$$

Then $\mathbf{z}(4t_1) = \mathbf{z}(0)$.

Proof. Direct calculation gives

$$\mathbf{F}(S_x \mathbf{z}) = -S_x \mathbf{F}(\mathbf{z}), \quad \mathbf{F}(S_v \mathbf{z}) = -S_v \mathbf{F}(\mathbf{z}).$$

Uniqueness at the point fixed by S_x yields

$$\mathbf{z}(t_1 + s) = S_x \mathbf{z}(t_1 - s).$$

Thus

$$\mathbf{z}(2t_1) = S_x \mathbf{z}(0) = (0, v_0).$$

Applying the same reflection argument at $t = 2t_1$ with S_v gives

$$\mathbf{z}(3t_1) = (-x_1, 0), \quad \mathbf{z}(4t_1) = (0, -v_0) = \mathbf{z}(0).$$

Uniqueness makes the trajectory periodic.